

## CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Founded by Richard C. Cabot  
 Eric S. Rosenberg, M.D., *Editor*  
 David M. Dudzinski, M.D., Meridale V. Baggett, M.D., Daniel Restrepo, M.D.,  
 Dennis C. Sgroi, M.D., Jo-Anne O. Shepard, M.D., *Associate Editors*, Emily K. McDonald, *Production Editor*



## Case 20-2025: An 86-Year-Old Woman with Neck Swelling and Dysphagia

Matthew L. Russell, M.D.,<sup>1,2</sup> Javier M. Romero, M.D.,<sup>3,4</sup>  
 Christopher J. Kwolek, M.D.,<sup>5,6</sup> Kevin A. Raskin, M.D.,<sup>1,2,7,8</sup>  
 and Yin P. Hung, M.D., Ph.D.<sup>9,10</sup>

## PRESENTATION OF CASE

Author affiliations are listed at the end of the article.

N Engl J Med 2025;393:290-300.

DOI: 10.1056/NEJMcpc2412527

Copyright © 2025 Massachusetts Medical Society.

*Dr. Emmett Kistler (Medicine):* An 86-year-old woman was evaluated at this hospital because of new neck swelling and dysphagia.

The patient had been in her usual state of health, which included end-stage kidney disease (ESKD) and dependence on dialysis, until 6 weeks before the current evaluation, when she was admitted to this hospital for dyspnea after missing 2 days of medications. During her hospitalization, she was found to have pulmonary edema and elevated systolic blood pressure (>200 mm Hg). She was treated with intravenous vasodilator therapy and supplemental sessions of hemodialysis; after 3 days, she was discharged with a prescription for oral antihypertensive medications.

After discharge, the patient received twice-weekly home visits by a nurse. One week before the current evaluation, she had a routine follow-up visit with her primary care physician. The oral temperature was 36.4°C, and the blood pressure 167/55 mm Hg; no other abnormalities were reported on examination.

Four days before the current evaluation, during a home visit, the patient reported new pain in the right side of the jaw and throat, which was worse with swallowing and radiated to her head, right eye, and right upper back. The oral temperature was 37.7°C, the heart rate 68 beats per minute, and the blood pressure 124/44 mm Hg. The application of a topical cooling agent to the jaw was recommended. During the next 2 days, jaw pain persisted; she could eat only soft foods and could drink minimal amounts of liquid. The patient was evaluated by a dentist, who reportedly found no evidence of dental infection. The next day, after hemodialysis, the patient presented to the emergency department of this hospital because of persistent jaw and throat symptoms.

The patient reported ongoing jaw pain, odynophagia, and limited oral intake during the past 4 days. She also noted pain in the right ear and on the right side of the head, as well as generalized fatigue and weakness. She had no history of neck trauma, surgery, or radiation therapy. She had had intermittent yellow discoloration of the saliva and weight loss of 3 kg during the past year. A review of systems was notable for right knee pain during the past week, which had led to

loss of ambulation; she had previously had no difficulty with ambulation. The patient wore dentures and reported no related problems. She produced a small amount of urine daily. She reported no fever, chills, dyspnea, cough, voice change, chest pain, palpitations, abdominal pain, nausea, change in bowel habits, vision or hearing change, or focal neurologic symptoms. She had no history of cognitive decline and was independent with respect to all basic activities of daily living; however, she needed assistance with three intermediate activities of daily living (housekeeping, laundry, and transportation).

The patient's medical history was notable for renal-cell carcinoma that had been treated with left nephrectomy; hypertensive nephropathy and secondary focal segmental glomerulosclerosis involving the right kidney, with ESKD, for which she received hemodialysis three times weekly; coronary artery disease for which she had undergone percutaneous coronary intervention; peripheral artery disease; mild asymptomatic carotid stenosis; hypertension; dyslipidemia; mild aortic stenosis; pulmonary hypertension; a patent foramen ovale; toxic multinodular goiter that had been treated with radioactive iodine ablation; a 9-cm soft-tissue mass in the right thigh that was thought to be an atypical lipoma; gastroesophageal reflux disease; cholelithiasis; chronic hepatitis C virus infection; thrombocytopenia; and cataract. Medications included aspirin, simvastatin, labetalol, lisinopril, nifedipine, omeprazole, vitamin supplementation designed for patients with kidney disease, calcium acetate, and as-needed acetaminophen, zolpidem, and oxycodone. Sulfa drugs had caused rash, and amlodipine had caused edema.

The patient was born and raised in East Asia and had lived in the western United States for most of her life. She was widowed and, during the past decade, had lived with relatives in the Boston area. She reported no other travel, and she had no known sick contacts. She had smoked one pack of cigarettes per day for 30 years but had quit three decades before the current evaluation; there was no other history of substance use. Her family history included cervical cancer in her mother and renal cancer in her father.

On examination, the temporal temperature was 36.7°C, the heart rate 78 beats per minute, the blood pressure 199/79 mm Hg, and the oxygen

saturation 95% while the patient was breathing ambient air. A large, tender mass was noted in the right submandibular area, without overlying skin changes, fluctuance, drainage, or warmth. All the teeth were missing except for the canine tooth of the right mandible; no lesions were observed in the oropharynx, and the mucous membranes were dry. Auscultation revealed a systolic ejection murmur (grade 2/6) at the base of the heart and carotid bruits that were more pronounced in the right artery than in the left artery. A thrill and bruit were detected over the fistula in the right arm. The right knee was tender and warm, and a small effusion was present; there was decreased range of motion owing to pain, but no erythema was noted. The right upper leg was enlarged and firm as compared with the left upper leg, but it was not tender or erythematous. No rash was present, and the remainder of the examination was normal.

The blood levels of calcium, magnesium, and phosphorus were normal; other laboratory test results are shown in Table 1. An electrocardiogram was normal.

*Dr. Javier M. Romero:* A chest radiograph showed mild pulmonary edema, trace bilateral pleural effusions, and aortic atherosclerosis. Radiographs of the right knee and right femur (Fig. 1A and 1B) showed mild osteopenia without fracture or dislocation.

*Dr. Kistler:* An otolaryngologic examination was notable for firm, markedly tender swelling of the right side of the neck. Fiberoptic laryngoscopy showed mild edema in the right lateral portion of the oropharynx and in the right arytenoid region, as well as erythema and thickening of the epiglottis. Treatment with intravenous ampicillin-sulbactam and dexamethasone was administered. Later that evening, the oxygen saturation was 89% while the patient was breathing ambient air. Supplemental oxygen was administered through a nasal cannula at a rate of 2 liters per minute.

*Dr. Romero:* Computed tomography (CT) of the right femur (Fig. 1C) revealed a small joint effusion in the right knee. The anterior thigh mass, which was observed to be composed primarily of fat, had markedly increased in size as compared with that seen on magnetic resonance imaging (MRI) of the femur performed 10 years earlier. The mass currently measured 13 cm by 10 cm by 25 cm and showed internal complexity.

**Table 1. Laboratory Data.\***

| Variable                               | Reference Range, Adults† | On Evaluation | Second Hospital Day | Late on Second Hospital Day |
|----------------------------------------|--------------------------|---------------|---------------------|-----------------------------|
| Sodium (mmol/liter)                    | 135–145                  | 140           | 140                 | 138                         |
| Potassium (mmol/liter)                 | 3.4–5.0                  | 3.2           | 4.3                 | 4.2                         |
| Chloride (mmol/liter)                  | 98–108                   | 96            | 96                  | 95                          |
| Carbon dioxide (mmol/liter)            | 23–32                    | 27            | 20                  | 21                          |
| Urea nitrogen (mg/dl)                  | 8–25                     | 6             | 23                  | 35                          |
| Creatinine (mg/dl)                     | 0.60–1.50                | 2.05          | 4.06                | 4.79                        |
| Glucose (mg/dl)                        | 70–110                   | 90            | 102                 | 121                         |
| Hemoglobin (g/dl)                      | 12.0–16.0                | 10.3          | —                   | 9.8                         |
| Hematocrit (%)                         | 36.0–46.0                | 30.9          | —                   | 30.2                        |
| White-cell count (per $\mu$ l)         | 4500–11,000              | 11,540        | —                   | 16,690                      |
| Differential count (per $\mu$ l)       |                          |               |                     |                             |
| Neutrophils                            | 1800–7700                | 9470          | —                   | 15,290                      |
| Lymphocytes                            | 1000–4800                | 940           | —                   | 820                         |
| Monocytes                              | 200–1200                 | 107           | —                   | 480                         |
| Platelet count (per $\mu$ l)           | 150,000–400,000          | 246,000       | —                   | 274,000                     |
| C-reactive protein (mg/liter)          | <8.0                     | —             | 219.1               | —                           |
| Erythrocyte sedimentation rate (mm/hr) | 0–20                     | —             | 99                  | —                           |

\* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

Ultrasonography of the neck (Fig. 1D) ruled out the presence of a clot in the right internal jugular vein; a focal irregular outpouching of the distal right common carotid artery that measured 14 mm in diameter and that had bidirectional flow was noted and was thought to be a pseudoaneurysm. Heterogeneous echogenicity and soft-tissue edema surrounded the outpouching, features that were most likely a result of inflammation. In addition, there was calcified and hyperechoic plaque in the right common carotid artery that extended to the bifurcation.

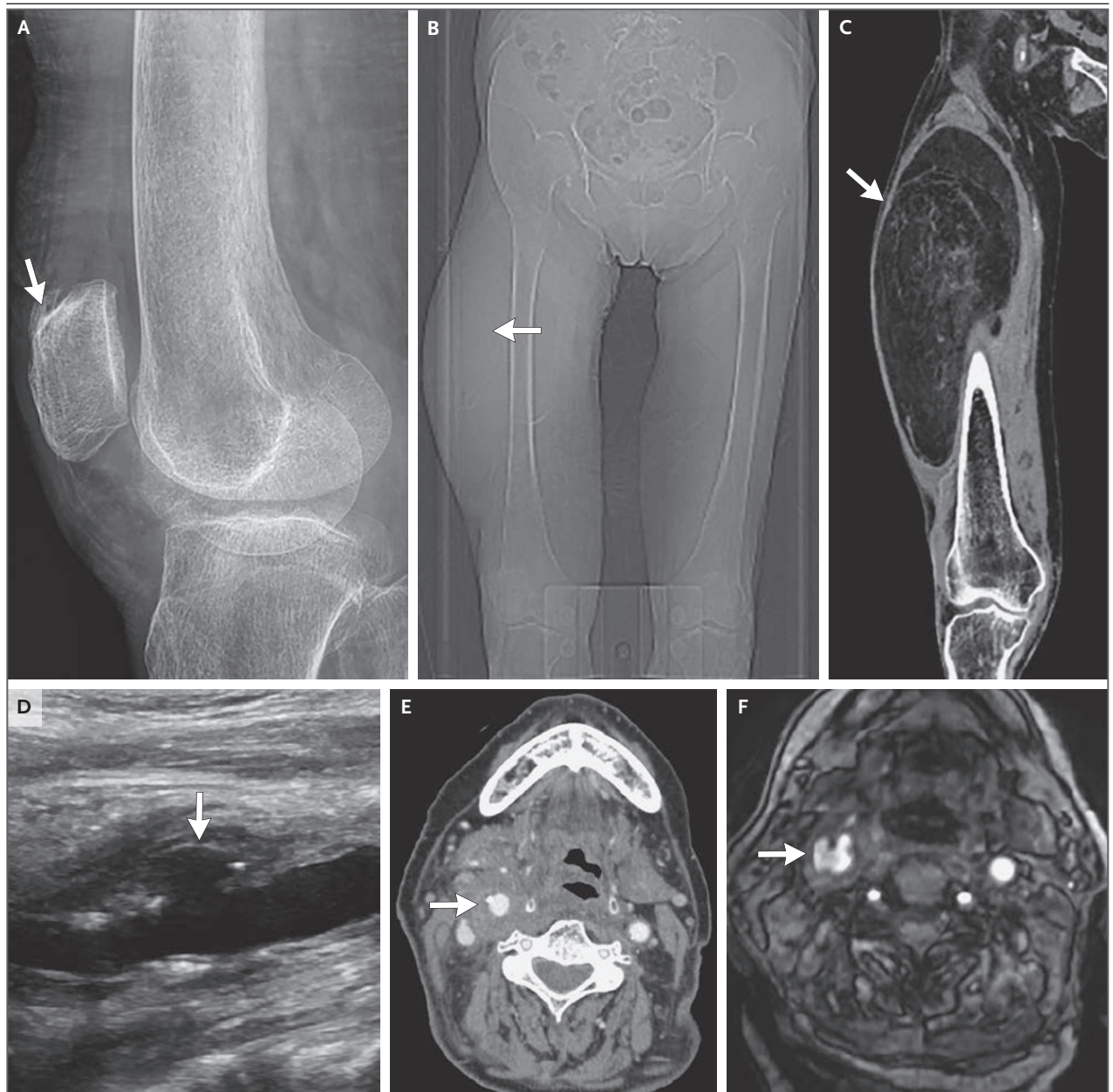
CT angiography of the neck, performed after the intravenous administration of contrast material (Fig. 1E), revealed extensive inflammation of the carotid space, with enhancement and thickening of the wall of the right carotid artery (including the common, internal, and external arteries) and asymmetric enlargement of the right palatine tonsil. These findings suggested the possibility of vasculitis, carotidynia, or an infectious process in the carotid space.

MRI and magnetic resonance angiography of

the neck (Fig. 1F) revealed focal dilatation and a 13-mm outpouching of the distal right common carotid artery with surrounding soft-tissue edema, findings that most likely indicated a pseudoaneurysm.

*Dr. Kistler:* On the second hospital day, specialists in infectious diseases, rheumatology, nephrology, dental medicine, and vascular and orthopedic surgery were consulted. Blood cultures were obtained. Dental examination showed gingival recession but no tooth pain, sensitivity, or motion. Laboratory test results are shown in Table 1; the blood level of C4 was normal, and a test for antinuclear antibodies (ANA) was positive at a titer of 1:40.

Arthrocentesis of the knee was performed, and 10 ml of cloudy fluid was obtained. Analysis of the fluid revealed 3000 erythrocytes per microliter and 53,000 nucleated cells per microliter, of which 93% were neutrophils. Gram's staining of the fluid showed abundant neutrophils, rare mononuclear cells, and no organisms; no crystals were noted. Transthoracic echocardiography



**Figure 1. Initial Imaging Studies.**

Imaging studies of the right knee, right femur, and neck were performed on admission. A lateral radiograph of the right knee (Panel A) shows no abnormalities aside from mild osteopenia (arrow). An anteroposterior radiograph of the right femur (Panel B) shows the previously detected large soft-tissue lesion in the vastus lateralis (arrow). A CT image of the right femur (Panel C) shows that the large anterior thigh mass (arrow) is composed primarily of fat. In addition, the mass shows some degree of internal septation and mild soft-tissue nodularity. An ultrasound image of the transverse right carotid artery (Panel D) shows a large luminal outpouching, which is most likely a pseudoaneurysm (arrow). An axial CT angiogram of the neck (Panel E) shows focal inflammation, wall thickening of the right carotid bulb, and fat stranding in the carotid space (arrow). An axial image from magnetic resonance angiography of the neck (Panel F) confirms the presence of the marked outpouching of the right common carotid artery, which is most likely a pseudoaneurysm (arrow).

showed normal ventricular function, a patent foramen ovale, and mild aortic stenosis, without valvular vegetation. Treatment with ampicillin–

sulbactam was stopped, and intravenous therapy with vancomycin and ceftriaxone was started.

Laboratory test results obtained late on the



second hospital day are shown in Table 1; the prothrombin time, prothrombin-time international normalized ratio, and partial-thromboplastin time were normal. The patient was admitted to the medicine unit of this hospital.

Diagnostic tests were performed, and additional management decisions were made.

#### DIFFERENTIAL DIAGNOSIS

*Dr. Matthew L. Russell:* Geriatric medicine is a field in which medical diagnostic and treatment decisions are heavily contextualized through the lens of a patient's expressed values, previous health care experiences, support systems, and prognosis. Most older adults have acquired various conditions and disease states owing to both genetic and lifestyle factors, as well as consequences from previous health care interventions.

It is important that we name at the outset the domains across which a geriatrician might be-

gin to frame diagnostic processes. A commonly used model to describe this approach is the age-friendly framework<sup>1</sup> of four evidence-based elements of care, known as "the 4Ms": mentation, medication, mobility, and what matters (Table 2).

#### THE PATIENT'S CHIEF CONCERN

The reason that this patient came to the hospital needs to be considered throughout her stay. In an 86-year-old person with many medical conditions, previously unknown issues may be identified that are neither the reason that she presented nor relevant to her overall clinical trajectory.

This is the second hospitalization in 2 months for this 86-year-old woman who is dependent on hemodialysis. She has swelling and tenderness on the right side of the neck and jaw, leading to dysphagia and odynophagia, as well as leukocytosis and evidence on imaging of a pseudoaneurysm with surrounding inflammation. She is also

**Table 2. The 4Ms Framework for the Care of Older Adults.<sup>2</sup>**

| Geriatric Domain                                                 | General Considerations                                                                                                                                                                                                                                    | Factors to Consider in This Patient                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mentation: addressing cognitive function and mood                | Screen for and treat dementia, depression, and delirium; these factors may affect decision-making capacity                                                                                                                                                | Whether moving in with family members had resulted from family's concerns about patient's cognitive impairment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Medication: reducing adverse drug effects                        | Assess for polypharmacy and potentially inappropriate medications<br>Ensure appropriate doses are being prescribed<br>Assess adherence                                                                                                                    | Assess for cognitive, financial, and dexterity-related barriers to medication adherence<br>Assess for appropriate timing of patient's routine oral medications in relation to scheduled hemodialysis sessions to ensure that medications are not lost during hemodialysis <sup>3</sup><br>Consider need for continued use of proton-pump inhibitor and severity of gastroesophageal reflux disease; if use still deemed appropriate, consider change to as-needed use<br>Clarify indication for oxycodone and ensure appropriate dose, given that it is partially removed during hemodialysis <sup>3</sup><br>Consider eliminating use of zolpidem, given that use in older adults is associated with falls, confusion, and death <sup>4,6</sup> |
| Mobility: maximizing physical function and independence          | Assess global measures of activities of daily living, mobility, and muscle mass, which help to assess "clinical reserve"<br>Assess instrumental activities of daily living<br>Screen for frailty<br>Assess home safety and support<br>Assess risk of fall | Probable frailty<br>Need for scores on Short Physical Performance Battery and Clinical Frailty Scale, which are appropriate scales to use in patients receiving hemodialysis <sup>7</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| What matters: aligning care with patient-identified health goals | Assess patient and family understanding<br>Assess decision-making capacity<br>Discuss advance directives<br>Discuss prognosis                                                                                                                             | Life expectancy of 86-year-old patient receiving hemodialysis most likely 1 to 3 years <sup>8</sup><br>Need for assessment of medical decision-making capacity<br>Patient's wishes in light of chronic disease (end-stage kidney disease) and recent illness                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

noted to have new swelling in the right knee with impairment of ambulation and an elevated neutrophil count in the joint fluid, as well as a markedly enlarging soft-tissue mass in the right thigh that had been presumed to be a lipoma.

#### ENDOCRINE OR BONE METABOLISM DISORDER

One of the defining features of this patient's presentation is jaw pain, which prompts consideration of osteonecrosis of the jaw. This condition can occur in patients who are receiving antiresorptive therapy and is noted to occur with higher frequency in patients with ESKD. In this patient, CT angiography of the neck did not show evidence of osteosclerosis, osteolysis, periosteal inflammation, or sequestrum (dead bone that separates from healthy bone), a finding that would be expected if she had osteonecrosis of the jaw. Soft-tissue inflammation around the jaw can occur in the context of osteonecrosis, but the absence of other findings makes osteonecrosis a relatively unlikely diagnosis in this patient.<sup>9</sup> Calciphylaxis, a rare but severe disorder that occurs most often in patients with ESKD who are receiving phosphate binders,<sup>10</sup> is a possible diagnosis in this case, but it would typically be associated with painful skin lesions and vascular calcifications.

The patient has a history of toxic multinodular goiter, which was previously treated with radioactive iodine ablation. The neck mass is probably not related to this treatment, but secondary cancers associated with this therapy can arise. However, these are quite rare in relation to the typical level of radiation administered in ablation therapy.<sup>11</sup>

#### AUTOIMMUNE DISORDER

An ANA titer of 1:160 or higher is typically suggestive of an autoimmune process, but a high positive ANA titer alone is not diagnostic. The prevalence of ANA positivity tends to be higher among older adults than among younger adults and has not been found to correlate with survival or with levels of serum markers of inflammation.<sup>12</sup> In addition, the presence of circulating autoantibodies is more common among patients with ESKD who are receiving hemodialysis than among hospitalized patients who are not receiving hemodialysis or among healthy persons.<sup>13,14</sup> This patient's diagnosis of chronic hepatitis C virus infection, coupled with vascular findings

involving the carotid arteries, suggests the possibility of cryoglobulinemic vasculitis. However, no skin findings, such as purpura or livedo, or neurologic findings, such as sensory motor polyneuropathy, that would support this diagnosis are noted.<sup>15</sup> No cryoglobulin test results are provided in the case history, but the C4 level is normal. It may be reasonable to complete the workup for vasculitis with serum protein electrophoresis, serum immunofixation, and measurement of rheumatoid factor.

#### INFECTION

The patient has ESKD and an arteriovenous fistula. Hemodialysis necessitates frequent access to her bloodstream, which puts her at increased risk for bacteremia.<sup>16,17</sup> In combination with advanced age, ESKD and other coexisting conditions may result in attenuated immune responses. Thus, the absence of fever or profound leukocytosis on presentation may not be reassuring in this patient.<sup>18</sup> The specificity for a white-cell count to detect infections is poor in older hospitalized adults.<sup>19</sup>

The patient has a painful, enlarging neck mass consistent with an infectious process. Given that she also has an asymmetric enlarged tonsil, the presence of a peritonsillar abscess with penetration into deeper tissue should be considered.<sup>20</sup> Similarly, the co-occurrence of new knee pain prompts concern about septic arthritis. The patient has a known history of mild aortic stenosis and a murmur detected on examination. Infective endocarditis in patients with ESKD most commonly affects the mitral and aortic valves.<sup>21</sup> Overall, my clinical suspicion for hematogenous spread of infection and possible endocarditis is high.

Duke criteria<sup>22</sup> for the diagnosis of infective endocarditis include major and minor criteria. The two major criteria are positive blood cultures and evidence of endocardial involvement.<sup>21</sup> This patient underwent transthoracic echocardiography, which did not show any obvious valvular vegetations, but this alone does not rule out infective endocarditis. She does have a known patent foramen ovale, which can lead to right-to-left shunting and allow bacteria to bypass the pulmonary circulation and reach the systemic circulation. If clinical suspicion for infective endocarditis remains high, we should consider performing transesophageal echocar-

diography, which offers greater sensitivity than transthoracic echocardiography for detecting vegetations.<sup>23</sup> That being said, we must also pause to consider the risks of performing transesophageal echocardiography, or any other invasive procedure, particularly in a medically frail 86-year-old person.

#### VASCULAR DISEASE

The patient has a pseudoaneurysm of the carotid artery. Vascular function can be negatively affected in patients with ESKD. Impaired endothelial function due to depletion of nitric oxide, oxidative stress, and changes in vascular stiffness and compliance can all occur in patients receiving hemodialysis.<sup>24,25</sup> The presence of adjacent soft-tissue swelling and edema is consistent with progressive vascular compromise.

#### A RED HERRING?

The mass in this patient's right thigh has increased in size. Lipomas are typically slow growing, "doughy" in consistency, mobile, and nontender. On CT, lipomas appear homogeneous and hypoattenuated.<sup>26</sup> It is reassuring that this patient's mass is not erythematous, tender, or warm, but the firmness on examination and the complexity seen on CT suggests that this mass may not be an uncomplicated lipoma. The possibility of hematogenous seeding of the lipoma should be considered, but it is important to remember that the mass on the thigh is not the reason that she presented to the hospital. Further investigation of the thigh mass may be warranted, but in this older patient, prioritizing diagnostic testing for the specific concerns that resulted in this hospitalization is key.

Still, we must consider whether the neck mass, the septic joint, and the thigh mass are all part of the same process or whether they are unrelated. In the care of an older adult, it is often unclear whether multiple coexisting conditions are related. We must assess which aspect of the clinical picture demands our attention now. In this patient, the neck symptoms and findings appear to be the most pressing.

The patient's clinical team broadened intravenous antibiotic coverage as a salient first step. In addition to the workup I proposed above, I return to the contextualization that is so important in geriatric care. After we know the patient's overall clinical trajectory and values and

have obtained blood culture results, our next priority will be to assess and address the neck mass, which is the reason she came to the hospital. Consultation with a vascular specialist and possible regular imaging or other investigation is needed to further assess what is probably an infected pseudoaneurysm of the neck. I would also consider additional evaluation for underlying endocarditis.

#### DR. MATTHEW L. RUSSELL'S DIAGNOSES

Deep-neck infection of either an endovascular aneurysm or a pseudoaneurysm.

Septic or inflammatory arthritis of the knee.

#### DISCUSSION OF MANAGEMENT

*Dr. Christopher J. Kwolek:* From the perspective of a vascular surgeon, the patient's problem list is fairly typical of our aging patient population, but a geriatric approach emphasizes the importance of assessing baseline functional status when making clinical decisions — especially those involving invasive procedures. This patient had a previously identified calcified carotid plaque, but at the current evaluation, she had a 14-mm outpouching that was consistent with a pseudoaneurysm. Follow-up CT performed on the third hospital day showed an increase in the size of the pseudoaneurysm as compared with that seen 2 days earlier on MRI. The cause of the pseudoaneurysm was most likely infection. The mechanism of infection could be direct extension, perhaps related to the patent foramen ovale or the brachiocephalic arteriovenous fistula, although she had no previous trauma or skin puncture, evidence of peritonsillar abscess or odontogenic infection, or hematogenous seeding. Because of the enlarging and tender infectious process in the neck, we performed surgical resection of the pseudoaneurysm, with electroencephalographic monitoring.

Bypass grafting was performed with the use of a nonprosthetic conduit to minimize the risk of infectious complications. Because of a mismatch in size between the carotid artery and the great saphenous vein, a cryopreserved femoral-artery graft was used. Owing to the occurrence of electroencephalographic slowing with test

clamping of the right carotid artery, a temporary shunt was inserted through the bypass graft. Gross examination showed marked inflammation. The bifurcation of the right common carotid artery and carotid artery, including the superior thyroid and ascending pharyngeal branches, was resected, and an ulcerated plaque with penetration into the vessel wall was detected. The external carotid artery was ligated. The patient remained neurologically intact throughout the operation.

#### PATHOLOGICAL DISCUSSION

*Dr. Yin P. Hung:* Histopathological examination of tissue obtained from the right carotid artery revealed abscess formation (Fig. 2A) and prominent acute inflammation with conspicuous neutrophils (Fig. 2B). Clusters of gram-positive cocci were highlighted by Brown–Hopps staining (Fig. 2C), and a concurrent wound culture grew *Staphylococcus aureus*. The pathological finding was indicative of infection of a pseudoaneurysm by *S. aureus*.

#### PATHOLOGICAL DIAGNOSIS

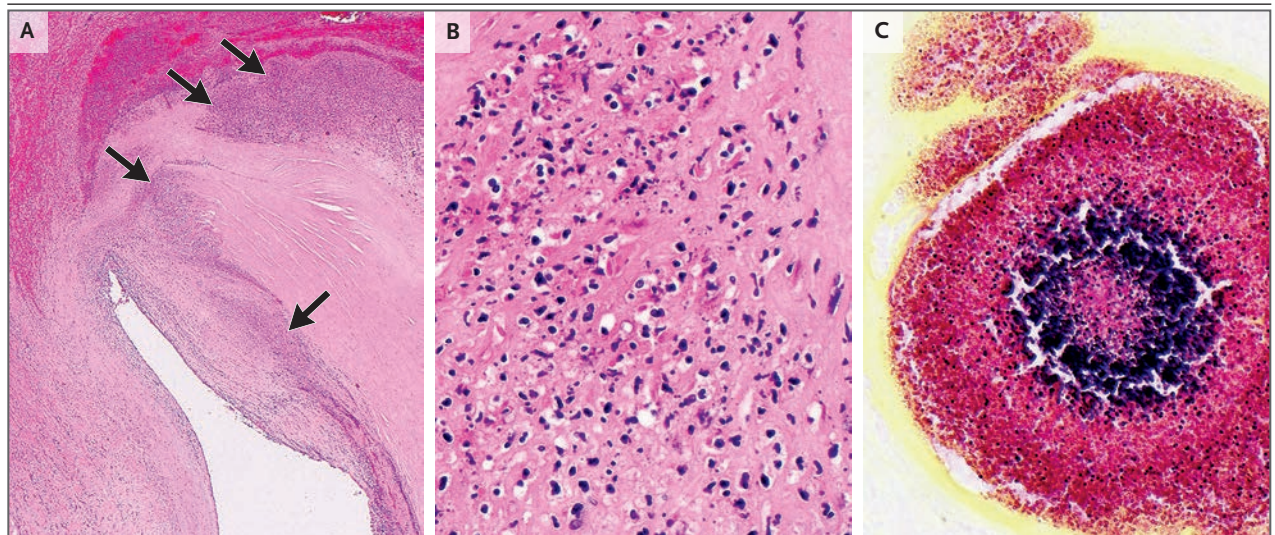
*Staphylococcus aureus* infection of a pseudoaneurysm involving the carotid artery.

#### HOSPITAL COURSE

*Dr. Kwolek:* After the surgical procedure, the patient underwent brief monitoring in the intensive care unit before discharge on postoperative day 4. Given that the cultures obtained during surgery grew methicillin-resistant *S. aureus*, the planned treatment at discharge was intravenous vancomycin, administered at the time of hemodialysis, for 6 weeks. During this time, the patient continued to live independently at home with her relatives. The patient's symptoms resolved, and on follow-up ultrasonography 10 weeks after discharge, the bypass graft of the right carotid artery appeared normal.

#### ORTHOPEDIC EVALUATION AND MANAGEMENT

*Dr. Kevin A. Raskin:* The orthopedic oncology service was consulted for surgical opinion and potential management of the patient's large thigh mass. The appearance of the mass on CT and the patient's clinical history were consistent with a benign lesion, which did not appear to be related to the current infection. However, 6 months after the current presentation, she noted that the mass was still painful and that the leg felt "tired" with walking.



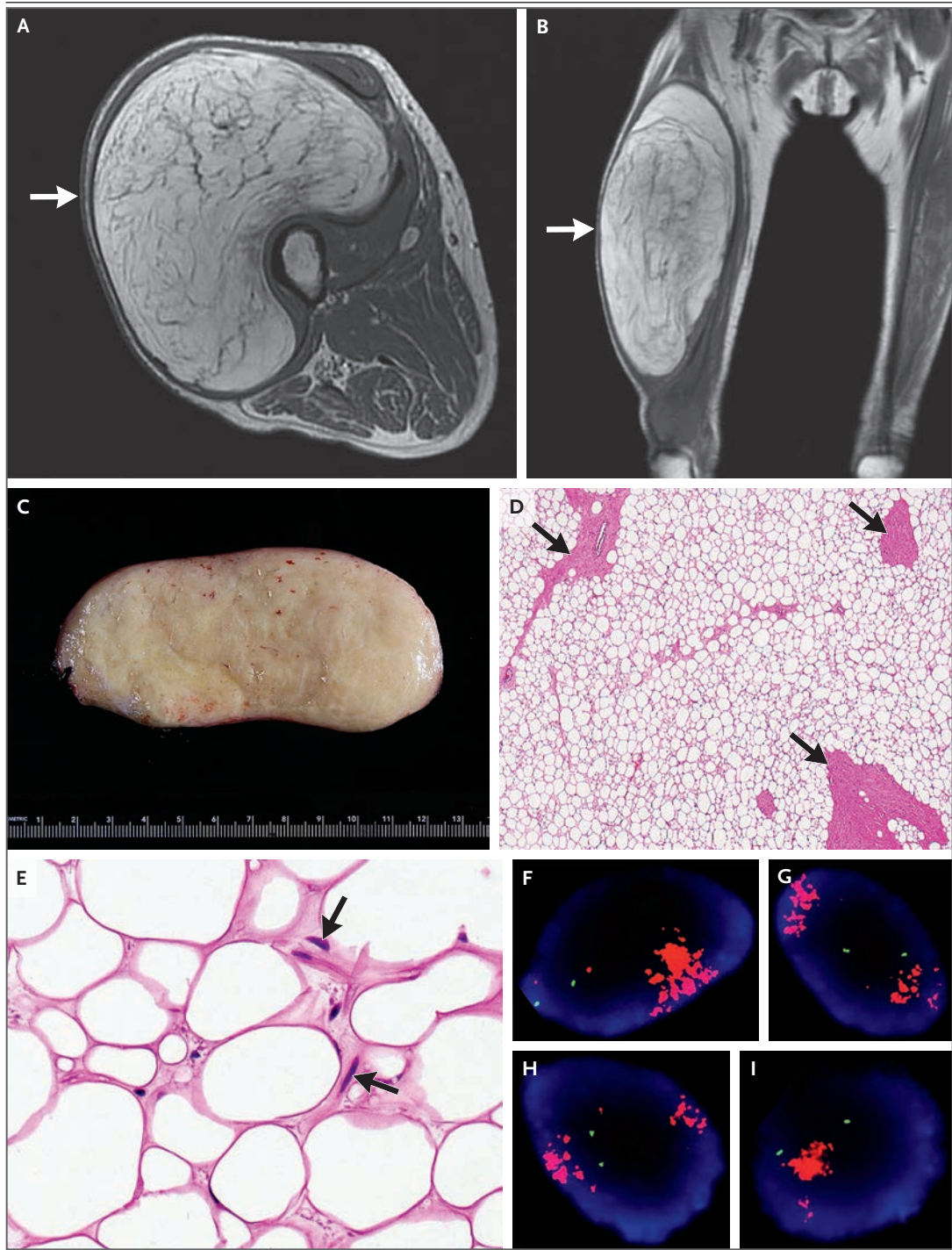
**Figure 2. Excision Specimen of the Right Carotid Artery.**

Hematoxylin and eosin staining of tissue obtained from the right carotid artery (Panel A) shows abscess formation with prominent acute inflammation (arrows). At higher magnification (Panel B), neutrophils are conspicuous. Brown–Hopps staining (Panel C) highlights clusters of gram-positive cocci.



*Dr. Romero:* An MRI of the femur was performed, and the findings were compared with those of the MRI of the femur that had been performed 10 years earlier. The new MRI revealed radiologically significant enlargement of

the intramuscular mass in the vastus lateralis (Fig. 3A and 3B). Although the mass was well-circumscribed and intrinsically hyperintense on T1-weighted imaging, with features consistent with the presence of fat, the number of septa-



tions had increased. A dominant septation separated the superior fourth portion of the mass, which showed increased signal on T2-weighted imaging that differed from the remainder of the mass. These findings prompted concern about the presence of an atypical lipomatous tumor such as a liposarcoma. In addition, the MRI showed a joint effusion at the right knee and partially visible degenerative changes at the right hip.

*Dr. Raskin:* Surgical removal of the mass was scheduled. Owing to the patient's need for hemodialysis, she was admitted to the hospital the day before her scheduled operation and underwent hemodialysis on the morning of her surgery. A marginal excision of the mass was performed with primary closure and the use of a closed-suction drain. The patient recovered uneventfully, and at a follow-up visit to the outpatient clinic 3 months after surgery, she reported that she had returned to full activity without symptoms.

#### PATHOLOGICAL AND CYTOGENETIC DISCUSSION

*Dr. Hung:* The patient's excised right-thigh mass was submitted in two large soft-tissue fragments, with one measuring 23 cm in greatest dimension and the other measuring 12 cm. On gross examination, a cross section of the mass had a homogeneous tan-yellow appearance (Fig. 3C). Microscopic examination of the tumor specimen

showed adipocytes that varied in size and had prominent collagenous fibrous septa (Fig. 3D). The nuclei were variably enlarged and hyperchromatic (Fig. 3E). Fluorescence in situ hybridization testing for *MDM2* amplification was positive (Fig. 3F through 3I), which confirmed the diagnosis of an atypical lipomatous tumor.

An atypical lipomatous tumor is a locally aggressive mesenchymal tumor composed of adipocytes. It is typically sporadic and frequently harbors amplifications of *MDM2* or *CDK4* (or both), which can be detected by means of fluorescence in situ hybridization testing or other methods. If dedifferentiation does not occur, there is minimal risk of metastasis. In contrast with lesions in the retroperitoneum, mediastinum, and paratesticular region that have the same histologic features, which are referred to as well-differentiated liposarcomas, the term atypical lipomatous tumors is used to describe lesions in the trunk or in the extremities (as in this patient), for which complete surgical resection would be curative.

#### FOLLOW-UP

*Dr. Kistler:* The patient recovered well from both surgical procedures and lived independently with her family. Over the course of the ensuing years, she had more frequent admissions, including hospitalization for heart failure and hypertensive emergency, a fall, and bleeding associated with the arteriovenous fistula. Four years after the surgical procedures, and 5 years after the current evaluation, the patient presented with weakness, cough, and dyspnea. Although she had been vaccinated against severe acute respiratory syndrome coronavirus 2, she became critically ill with coronavirus disease 2019. She was transitioned to comfort care and died peacefully.

#### FINAL DIAGNOSES

*Staphylococcus aureus* infection of a pseudoaneurysm involving the carotid artery.

Atypical lipomatous tumor of the thigh.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank Dr. Cara A. Lund for her assistance with the case narrative.

#### Figure 3 (facing page). MRI of the Right Leg and Excision Specimen of the Right-Thigh Mass.

MRI was performed 6 months after the current presentation. Axial and coronal images of the right leg (Panels A and B, respectively) show a soft-tissue mass (arrows) in the lateral soft tissues of the right thigh, within the vastus lateralis, that appears to be composed predominantly of fat. In addition, internal septations show mild enhancement. Surgical removal of the mass was performed 1 month later. A cross section of the excised mass (Panel C) has a homogeneous tan-yellow appearance. Hematoxylin and eosin staining of the tissue specimen (Panel D) shows adipocytes with prominent fibrous septa (arrows). At higher magnification (Panel E), the nuclei appear enlarged and hyperchromatic (arrows). Fluorescence in situ hybridization testing (Panels F through I) for *MDM2* amplification is positive (red signal, *MDM2*; green signal, centromeric region of chromosome 12).

## AUTHOR INFORMATION

<sup>1</sup>Department of Medicine, Massachusetts General Hospital, Boston; <sup>2</sup>Department of Medicine, Harvard Medical School, Boston; <sup>3</sup>Department of Radiology, Massachusetts General Hospital, Boston; <sup>4</sup>Department of Radiology, Harvard Medical School, Boston; <sup>5</sup>Department of Vascular Surgery, Newton–

Wellesley Hospital, Newton, MA; <sup>6</sup>Department of Surgery, Harvard Medical School, Boston; <sup>7</sup>Department of Orthopaedic Surgery, Massachusetts General Hospital, Boston; <sup>8</sup>Department of Orthopaedic Surgery, Harvard Medical School, Boston; <sup>9</sup>Department of Pathology, Massachusetts General Hospital, Boston; <sup>10</sup>Department of Pathology, Harvard Medical School, Boston.

## REFERENCES

1. Institute for Healthcare Improvement. Age-friendly health systems (<https://www.ihi.org/networks/initiatives/age-friendly-health-systems>).
2. Mate K, Fulmer T, Pelton L, et al. Evidence for the 4Ms: interactions and outcomes across the care continuum. *J Aging Health* 2021;33:469-81.
3. Leuppi-Taegtmeier A, Duthaler U, Hammann F, et al. Pharmacokinetics of oxycodone/naloxone and its metabolites in patients with end-stage renal disease during and between haemodialysis sessions. *Nephrol Dial Transplant* 2019;34:692-702.
4. Lai M-M, Lin C-C, Lin C-C, Liu C-S, Li T-C, Kao C-H. Long-term use of zolpidem increases the risk of major injury: a population-based cohort study. *Mayo Clin Proc* 2014;89:589-94.
5. Westermeyer J, Carr TM. Zolpidem-associated consequences: an updated literature review with case reports. *J Nerv Ment Dis* 2020;208:28-32.
6. Finkle WD, Der JS, Greenland S, et al. Risk of fractures requiring hospitalization after an initial prescription for zolpidem, alprazolam, lorazepam, or diazepam in older adults. *J Am Geriatr Soc* 2011;59:1883-90.
7. Imamura K, Yamamoto S, Suzuki Y, et al. Comparison of the association between six different frailty scales and clinical events in patients on hemodialysis. *Nephrol Dial Transplant* 2023;38:455-62.
8. Sladoje-Martinovic B, Mikolasevic I, Bubic I, Racki S, Orlic L. Survival of chronic hemodialysis patients over 80 years of age. *Clin Interv Aging* 2014;9:689-96.
9. Obinata K, Shirai S, Ito H, et al. Image findings of bisphosphonate related osteonecrosis of jaws comparing with osteoradionecrosis. *Dentomaxillofac Radiol* 2017;46:20160281.
10. Case records of the Massachusetts General Hospital (Case 31-2001). *N Engl J Med* 2001;345:1119-24.
11. Haugen BR, Alexander EK, Bible KC, et al. 2015 American Thyroid Association management guidelines for adult patients with thyroid nodules and differentiated thyroid cancer: the american thyroid association guidelines task force on thyroid nodules and differentiated thyroid cancer. *Thyroid* 2016;26:1-133.
12. Hurme M, Korkki S, Lehtimäki T, et al. Autoimmunity and longevity: presence of antinuclear antibodies is not associated with the rate of inflammation or mortality in nonagenarians. *Mech Ageing Dev* 2007;128:407-8.
13. Nolph KD, Husted FC, Sharp GC, Siemsen AW. Antibodies to nuclear antigens in patients undergoing long-term hemodialysis. *Am J Med* 1976;60:673-6.
14. Undas A, Kolarz M, Kopec G, Glowacki R, Placzekiewicz-Jankowska E, Tracz W. Autoantibodies against N-homocysteinylation proteins in patients on long-term hemodialysis. *Nephrol Dial Transplant* 2007;22:1685-9.
15. Cacoub P, Vieira M, Saadoun D. Cryoglobulinemia — one name for two diseases. *N Engl J Med* 2024;391:1426-39.
16. Lok CE, Huber TS, Lee T, et al. KDOQI clinical practice guideline for vascular access: 2019 update. *Am J Kidney Dis* 2020;75:Suppl 2:S1-S164.
17. Locham S, Naazie I, Canner J, Siracuse J, Al-Nouri O, Malas M. Incidence and risk factors of sepsis in hemodialysis patients in the United States. *J Vasc Surg* 2021;73(3):1016-1021.e3.
18. High KP, Bradley SF, Gravenstein S, et al. Clinical practice guideline for the evaluation of fever and infection in older adult residents of long-term care facilities: 2008 update by the Infectious Diseases Society of America. *J Am Geriatr Soc* 2009;57:375-94.
19. Compté N, Dumont L, Bron D, et al. White blood cell counts in a geriatric hospitalized population: a poor diagnostic marker of infection. *Exp Gerontol* 2018;114:87-92.
20. Galioto NJ. Peritonsillar abscess. *Am Fam Physician* 2017;95:501-6.
21. Vasudev R, Shah P, Kaur S, et al. Infective endocarditis in hemodialysis patients. *J Heart Valve Dis* 2016;25:369-74.
22. Durack DT, Lukes AS, Bright DK, Duke Endocarditis Service. New criteria for diagnosis of infective endocarditis: utilization of specific echocardiographic findings. *Am J Med* 1994;96:200-9.
23. Otto CM, Nishimura RA, Bonow RO, et al. 2020 ACC/AHA guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Thorac Cardiovasc Surg* 2021;162(2):e183-e353.
24. Meyer C, Heiss C, Drexhage C, et al. Hemodialysis-induced release of hemoglobin limits nitric oxide bioavailability and impairs vascular function. *J Am Coll Cardiol* 2010;55:454-9.
25. Jourde-Chiche N, Dou L, Cerini C, Dignat-George F, Brunet P. Vascular incompetence in dialysis patients — protein-bound uremic toxins and endothelial dysfunction. *Semin Dial* 2011;24:327-37.
26. Derin AT, Yaprak N. Lipomas: review and evaluation of the literature. *Clin Surg* 2017;2:1615 (<https://www.clinicsinsurgery.com/open-access/lipomas-review-and-evaluation-of-the-literature-3049.pdf>).

Copyright © 2025 Massachusetts Medical Society.